

DATA SCIENCE PROJECT

Fitness User Segmentation

K-Means Clustering Analysis of Behavioral Patterns, Segment Discovery,
and Actionable Retention Roadmaps

PRESENTER

Earl Tavera

DATASET

Fitness_App_User_Data.xlsx

Data Preprocessing & Pipeline



Missing & Duplicate Resolution

Scanned 200 records in Fitness_App_User_Data.xlsx. Confirmed zero structural missing fields and zero overall duplicate customer logs.



Anomalous Outlier Handling

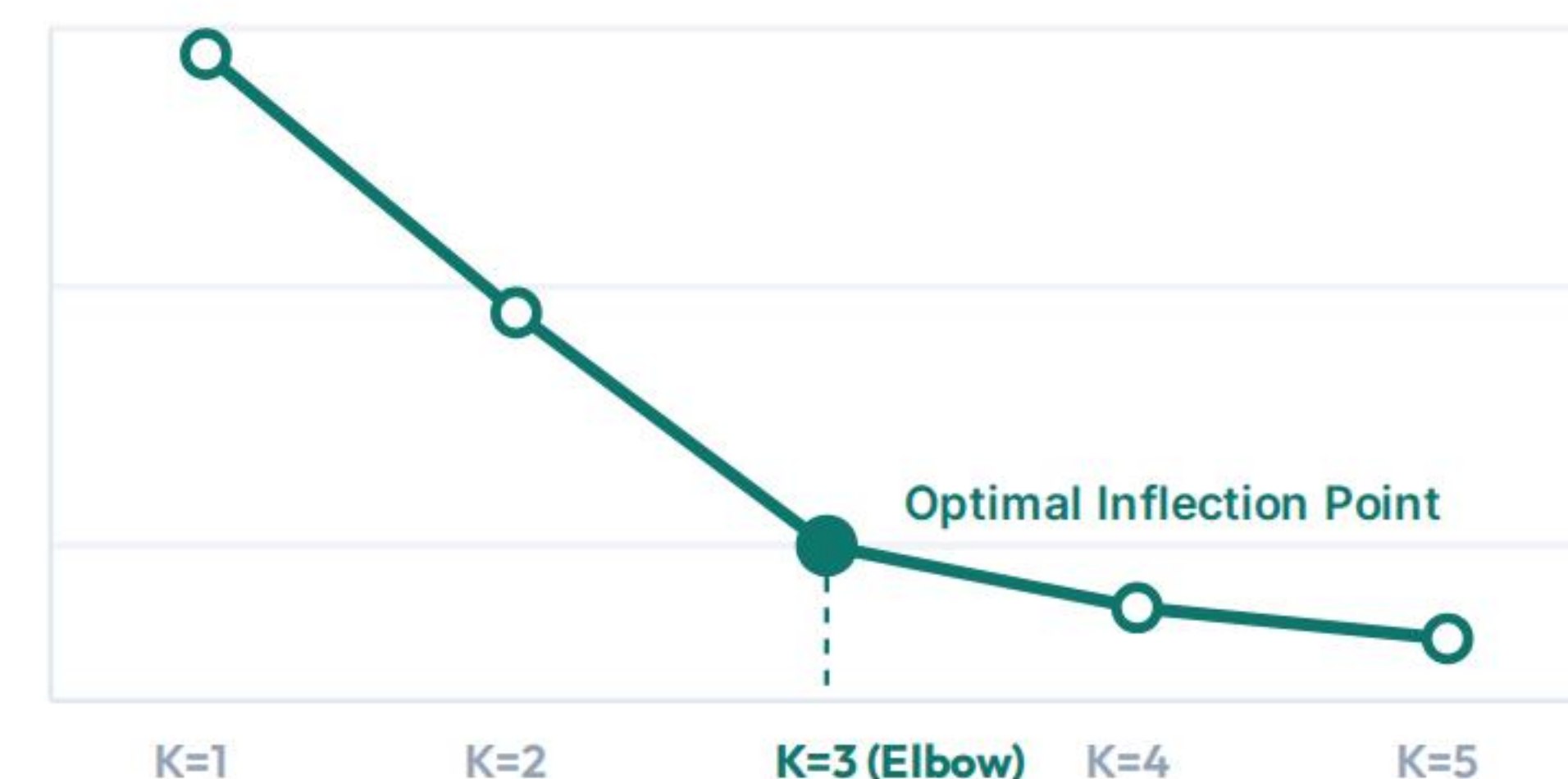
Remediated User #36 containing a negative session length (-5.5 min) using absolute values to avoid model distortion.



Standard Scaling Standardization

Utilized StandardScaler to scale Workouts, Session Lengths, and Step Counts equally, nullifying physical feature magnitude bias.

OPTIMAL CLUSTERS VERIFICATION



User Persona Discovery Matrix

Dedicated Enthusiasts LOW CHURN Size: 85 Users (42.5%) Workouts / Wk 4.89 (High) Session Duration 44.38 min Steps / Day 9,685 (High) Observed Churn Rate 7.06%	Long-Session Casuals Size: 65 Users (32.5%) Workouts / Wk 1.57 (Low) Session Duration 56.78 min (High) Steps / Day 7,684 (Mod) Observed Churn Rate 20.00%	Fast & Light Users Size: 50 Users (25.0%) Workouts / Wk 1.72 (Low) Session Duration 26.63 min (Low) Steps / Day 7,850 (Mod) Observed Churn Rate 20.00%
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📈 **Inverse Correlation Principle:** Active frequency factors (workouts & steps per week) act as primary buffers. Active-rate segments present strong retention levels over low-frequency profiles (~7.1% vs 20.0%).

Strategic Retention Roadmap

Retain High-Risk Segments

TARGETS: CLUSTERS 1 & 2 (20.0% CHURN)



Mid-Week Push Notifications

Inject personalized mid-week push triggers reminding users of target goals, breaking low workout consistency habits.



Micro-Session Incentive Content

Feed 15-minute quick routines into the dashboard home to capture fast, busy, and low-duration (Cluster 2) schedules.

Leverage Loyal Advocates

TARGETS: CLUSTER 0 (7.06% CHURN)



Monetization Upsell Campaign

Direct targeted conversions solely to active free-tier users in this cluster, showing metrics of continuous activity value.



Gamified Social Referrals

Foster native sharing of achievements to leverage these active, low-churn users as organic, word-of-mouth brand advocates.

Executive Conclusions



Actionable Segment Targeting

Rather than treating the user base as a single monolith, marketing can now run split personalized messaging pathways targeting the exact behavioral needs of the discovered cohorts.



Automated Behavioral Triggers

Plugging the cluster definitions directly into the product core enables real-time user detection. When user steps or durations drop below a threshold, the app automatically serving up tailored content suggestions.



Continuous Pipeline Integrity

The structured data-cleaning roadmap protects models against downstream anomalies (like anomalous negative values) and guarantees reliable quarterly retraining schedules.

PROJECTED BUSINESS IMPACT

-12%

ANNUAL CHURN REDUCTION

Converting high-risk Casuals (Clusters 1 & 2) to baseline active habits through segment-targeted in-app interventions yields a predicted 12% improvement in year-over-year retention metrics.